
Diffuse and Steer: Corrective Sampling for Stable 3D Molecular Diffusion

Fang Wan¹, Wenhan Gao², Jingxiang Qu¹, Yi Liu^{1,2*}

¹ Department of Computer Science, Stony Brook University

² Department of Applied Mathematics and Statistics, Stony Brook University

Abstract

Diffusion models have achieved state-of-the-art performance across diverse domains, yet their application to molecular generation remains challenging. Unlike many data types where values can tolerate slight variations, such as pixel intensities in images, molecules are governed by strict geometric and chemical constraints: minor variations in the atomic coordinates of even a single atom can lead to totally invalid or unstable molecules. These constraints give rise to *highly concentrated* data distributions, forming sharp probability peaks. Moreover, these peaks are *densely packed* in configuration space: changing one atom’s type, along with small but precise adjustments to its position and that of its neighbors, can result in a distinct molecule, whereas images generally require much larger perturbations to change semantic meaning. This dense-concentrated structure makes diffusion modeling fragile: because valid regions are narrow and tightly clustered, even small deviations at intermediate timesteps can easily cross validity boundaries. Once entering the invalid regions, the generative process provides unreliable guidance, causing errors that accumulate over timesteps and drift generative trajectories off-distribution, ultimately leading to irreparable structural violations. To address this challenge, we formalize the notion of dense-concentrated structure in molecular distributions and analyze how discrepancies at intermediate steps propagate under reverse inference. Building on this insight, we propose **DIST**, a plug-in corrective method that **D**iffuses and **S**TEers the intermediate distribution, thereby realigning inference trajectories toward a valid molecular distribution. Our method is model-agnostic and can be integrated into a wide range of existing diffusion models, achieving significant improvements in performance while reducing the computational cost to nearly half the standard number of timesteps. The project is available at <https://github.com/jiuguaiwf/DIST>.

1 Introduction

Generative models are probabilistic frameworks that aim to approximate an underlying data distribution and generate new samples from the learned distribution. By providing a principled approach to learning and sampling from complex, high-dimensional distributions, generative modeling has emerged as a promising paradigm with broad implications for design automation, simulation, and scientific discovery. Recently, diffusion models (DMs) [Ho et al., 2020, Song et al., 2021b] have become a prominent generative paradigm due to their outstanding performance in natural image synthesis and beyond [Song et al., 2020, Rombach et al., 2022, Watson et al., 2023]. A DM consists of a forward process and a reverse process. In the forward process, data samples are gradually corrupted by a Markovian noise injection until they become indistinguishable from pure Gaussian noise. The reverse process is parameterized by a neural network, which is trained to approximate the time-reversed dynamics by iteratively denoising the corrupted states. At inference time, the model generates new samples by simulating this learned reverse trajectory, reconstructing structured

*Correspondence to yi.liu.4@stonybrook.edu

data from pure noise. Recent work has extended DMs to 3D molecular generation [Hoogeboom et al., 2022, Xu et al., 2023]. However, molecular data presents unique challenges that make direct application of diffusion models less effective.

Specifically, 3D molecules are represented by continuous 3D atomic coordinates together with discrete features such as atom types. Unlike images, where pixel intensities are only loosely correlated and can tolerate a wide range of variations, molecules are governed by strict geometric and chemical constraints, such that even small perturbations to atomic coordinates or atom types can result in completely invalid or unstable structures [Choi et al., 2025]. These constraints result in highly concentrated data distributions with narrow probability peaks, where each peak represents a valid and stable molecular configuration. Even slight displacements can shift the molecular configuration off-peaks into regions of negligible probability, corresponding to invalid or unstable states [Reymond et al., 2012, Martin and Cao, 2015, Bohde et al., 2025]. Moreover, these peaks are densely packed but clearly separated: changes in one atom’s type, along with small (densely packed) but precise adjustments (well separated) to its position and that of its neighbors, can result in a distinct molecule. **Overall, the molecular distribution exhibits an evident dense and concentrated structure, where each probability peak corresponds to a chemically valid molecule, and the regions between the peaks are of near-zero density.** We provide an illustrative analogy to compare the distribution and diffusion process of images with those of molecules in Fig. 1, to highlight the consequences of such a dense and concentrated structure to the diffusion process. Notably, such denseness breaks the clear supervision signal required for denoising, introduces learning difficulties, and leads to errors that accumulate over time; and because of the concentration of the molecular distribution, such errors cannot be tolerated, ultimately resulting in invalid and unstable generations.

Under the same forward noising process, the peaks of molecular distributions quickly merge creating overlap regions where samples become indistinguishable. In contrast, image distributions exhibit broader peaks that overlap smoothly and only at later stages. However, for the reverse process of molecular diffusion, a critical problem arises: **overlap regions create intersections or crossings of generative trajectories which make the score field inherently ambiguous**, where multiple plausible directions coexist, but the model can only represent a single *averaged* vector. As a result, the learned score is systematically inaccurate in these regions [Liu et al., 2023, Lee et al., 2023, Ni et al., 2025]. Because the peaks are thin, discretization error [Zhang et al., 2023], model limitations, and imperfect score estimation in overlap regions can push the reverse updates too far, placing samples into low-density regions (see Fig. 1). The resulting discrepancy between the true data distribution and the model distribution, caused by artificial inflation of probability mass in invalid regions, then accumulates and propagates [Li and van der Schaar, 2024], ultimately leading to irreversible structural failures. We further analyze this phenomenon in Sec. 3.1.

To address this challenge, we focus on the unique nature of molecular data distributions. Since chemically valid molecules occupy only the densely packed distribution peaks, which are confined to narrow and well-separated regions of the representation space, we describe this property as *dense-concentrated structure (DC-structure)*, formally introduced in Definition 3.1 in Sec. 3.1. This definition provides a quantitative handle on the geometry of molecular distributions and lays the theoretical foundation for our analysis. Building on this, we show in Sec. 3.2 how such analysis motivates a corrective method, **DIST**, which **D**iffuses the intermediate distribution and **S**Teers trajectories back toward valid high-density regions. DIST improves the stability and overall performance of molecular generation, while also providing efficiency gains as an additional benefit.

In this work, our main contributions are: (1) **Observation**. We are the first to highlight that molecular data distributions are highly *concentrated* and *dense*, making diffusion-based generation fragile; (2) **Theory**. We formalize DC-structure in molecular distributions and analyze its effects on intermediate distributions and error propagation in reverse inference; (3) **Method**. We design DIST, a training-free plug-in module that corrects the intermediate model marginal distribution rather than applying post-hoc filtering; and (4) **Performance**. Experiments on multiple benchmarks and backbones show that DIST improves stability and overall performance while reducing computational cost to nearly half the standard number of timesteps.

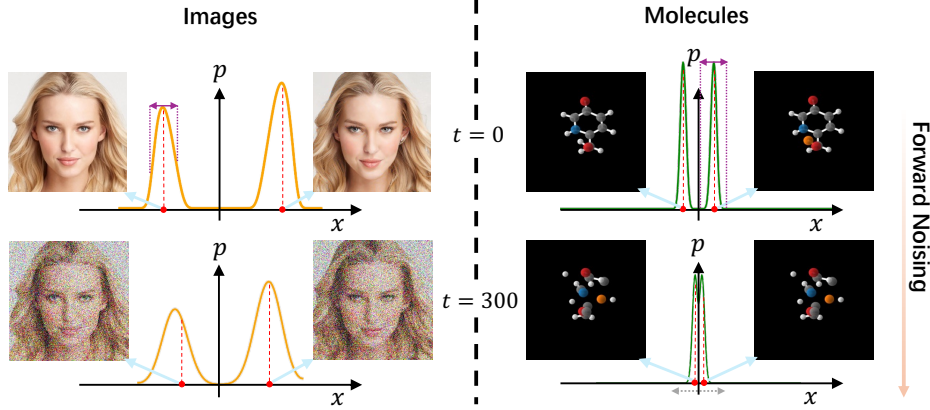


Figure 1: Diffusion process applied to pairs of images (left, obtained from Preechakul et al. [2022]) and molecules (right, extracted from QM9 [Ramakrishnan et al., 2014]) under the same noise schedule. Both pairs start from similar, yet distinct configurations, corresponding to two separate peaks in the distribution. Compared with the image distribution, the molecular distribution is much denser and more concentrated with narrower peaks. Under mild noise corruption at $t = 300$, noisy images remain distinguishable, whereas *noisy molecules quickly become indistinguishable due to the denseness*. At $t = 0$, small errors (indicated in purple) in the image distribution still land in regions of relatively high density, corresponding to visually realistic images, whereas *small errors in the molecular distribution will drift the generated samples into regions of near-zero density between two peaks due to concentration*.

2 Preliminaries and Related Work

2.1 Diffusion Models

Diffusion models (DMs) [Ho et al., 2020, Song et al., 2021b] are latent-variable generative models that learn to transform Gaussian noise into data samples through a forward–reverse Markov chain. Let $\mathbf{x} \sim p(\mathbf{x})$ denote a clean data sample, and $\mathbf{z}_t$ denote its progressively noised version at timestep $t \in \{0, \dots, T\}$. Here T is the total number of timesteps, $\beta_t \in (0, 1)$ is a variance-schedule parameter, and we use the shorthand $\mathbf{z}_{1:T} = (\mathbf{z}_1, \dots, \mathbf{z}_T)$. The forward process gradually corrupts data by adding Gaussian noise:

$$p(\mathbf{z}_{1:T} | \mathbf{x}) = \prod_{t=1}^T p(\mathbf{z}_t | \mathbf{z}_{t-1}), \quad p(\mathbf{z}_t | \mathbf{z}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \mathbf{z}_{t-1}, \beta_t I). \quad (1)$$

By composition, the marginal conditional distribution admits a closed form:

$$p(\mathbf{z}_t | \mathbf{x}) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}, (1 - \bar{\alpha}_t)I), \quad \bar{\alpha}_t = \prod_{s=1}^t \alpha_s = \prod_{s=1}^t (1 - \beta_s). \quad (2)$$

Here $\alpha_s = 1 - \beta_s$ controls the noising pace [Ho et al., 2020, Nichol and Dhariwal, 2021]. The unconditional marginal at step t is then

$$p(\mathbf{z}_t) = \int p(\mathbf{x}) \mathcal{N}(\mathbf{z}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}, (1 - \bar{\alpha}_t)I) d\mathbf{x}, \quad (3)$$

which interpolates between the data distribution $p(\mathbf{x})$ and the Gaussian prior $p(\mathbf{z}_T) \approx \mathcal{N}(0, I)$ [Albergo et al., 2025]. The reverse process reconstructs data from noise, factorized as $q_\theta(\mathbf{z}_{0:T}) = q(\mathbf{z}_T) \prod_{t=1}^T q_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t)$, with transitions $q_\theta(\mathbf{z}_{t-1} | \mathbf{z}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{z}_t, t), \rho_t^2 I)$, where $\boldsymbol{\mu}_\theta$ is predicted by a neural network and ρ_t is typically fixed. DMs are trained with the noise-prediction objective [Song et al., 2021b]:

$$\mathcal{L}_{\text{DM}} = \mathbb{E}_{\mathbf{x}, \varepsilon, t} [\|\varepsilon - \varepsilon_\theta(\mathbf{z}_t, t)\|^2], \quad \mathbf{z}_t = \sqrt{\bar{\alpha}_t} \mathbf{x} + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I). \quad (4)$$

The network ε_θ can be interpreted as learning the score field $\nabla_{\mathbf{z}_t} \log p(\mathbf{z}_t)$ [Song et al., 2021a,b]. New samples are generated by starting from pure Gaussian noise $\mathbf{z}_T \sim \mathcal{N}(0, I)$ and iteratively

applying the reverse update:

$$\mathbf{z}_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \boldsymbol{\varepsilon}_\theta(\mathbf{z}_t, t) \right) + \rho_t \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, I). \quad (5)$$

2.2 Diffusion Models for 3D Molecular Generation

A 3D molecule with N atoms comprises both continuous atomic coordinates and discrete atomic features [Hong et al., 2025]. The atomic coordinates are represented as $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N) \in \mathbb{R}^{N \times 3}$, where each $\mathbf{x}_i \in \mathbb{R}^3$ specifies the position of an atom in Euclidean space. Atomic features, such as atom types, are encoded as $\mathbf{h} = (\mathbf{h}_1, \dots, \mathbf{h}_N) \in \mathbb{R}^{N \times d}$. Consequently, molecular data exhibit an intrinsic hybrid discrete–continuous structure, which has motivated a range of complementary modeling paradigms, including equivariant denoising networks [Hoogeboom et al., 2022, Xu et al., 2023], canonicalization-based methods [Kaba et al., 2023], and latent-space diffusion approaches [Xu et al., 2023, Ding and Hofmann, 2025, Chen et al., 2025, Luo et al., 2025]. Beyond architectural design, numerous works have sought to improve molecular diffusion models from different perspectives, such as incorporating chemical priors [Huang et al., 2024] and enhancing geometric alignment [Ding and Hofmann, 2025]. A broader discussion of molecular generative models is provided in Appendix A.

Despite substantial progress, errors induced by the imperfectly learned score network (see Eq. (4)) remain pervasive and ignored in current molecular diffusion sampling [Song et al., 2023, 2024, Joshi et al., 2025]. We observe this phenomenon consistently across GNN- and Transformer-based architectures, as well as across both equivariant and non-equivariant formulations. As these errors accumulate over diffusion timesteps, the discrepancy between the true data marginal distribution and the model-induced distribution progressively increases during sampling. This observation suggests that **error accumulation cannot be fully mitigated by architectural or representational choices alone**. Rather, it underscores the importance of explicitly correcting inference trajectories at intermediate timesteps to curb error propagation and reduce distributional drift.

2.3 Relations with Prior Work

Related corrective diffusion methods mainly refine individual samples or stabilize local denoising transitions through guidance, candidate selection, or training- and schedule-level modifications [Papalampidi et al., 2025, Kim et al., 2025, Ning et al., 2024, Wang et al., 2025, Li et al., 2024a]. DIST instead targets the molecular-specific DC-structure and corrects the intermediate model distribution before the remaining reverse process. Although DIST is related to rejection/filtering at the workflow level [Von Neumann, 1951, Gordon et al., 1993], it shares computation up to the correction timestep and selects local regions rather than filtering only fully generated samples. Thus, DIST provides a training-free, plug-in, and efficient distribution-level correction for molecular diffusion. A detailed comparison with related corrective diffusion and rejection/filtering is provided in Appendix B.

3 Method

In this section, we delve into three key questions: (1) How can the unique structure of molecular distributions, constrained by chemical rules, be formally characterized? (2) What issues arise due to this structure for 3D molecular diffusion models? (3) Can these issues be mitigated through correction? We answer the first two questions by formally investigating the DC-structure of molecular distributions in Sec. 3.1. Building on this insight, we propose DIST together with its theoretical analysis in Sec. 3.2, which addresses the last question.

3.1 Dense-concentrated Structure Issue

As illustrated in Fig. 1, molecular data distribution over the representation space exhibits an evident DC-structure, where each peak corresponds to a chemically valid molecule, and regions between the peaks are of near-zero density. This contrasts with images, where the pixel values can tolerate a wide range of variations, resulting in wider peaks and smoother transitions. **To rigorously capture this phenomenon and further analyze its implications, we next formalize the DC-structure in probabilistic terms.** Consistent with Sec. 2 and prior work, we denote the true and model marginals by $p(\mathbf{z}_t)$ and $q_\theta(\mathbf{z}_t)$, respectively. Unless otherwise stated, all analysis in this work is carried out under the molecular setting rather than the universal diffusion machinery. For notational simplicity, we write the true marginal as p_t , omit the learnable parameter θ , and write the model marginal as q_t .

Definition 3.1 (Dense-concentrated Structure). *There exist K_0 centers $\{m_k\}$, a scale $\sigma_* > 0$, a separation $\Delta > 0$, and weights $\{w_k\}$ such that, for the operative noise level t ,*

$$p_t \simeq \sum_{k=1}^{K_0} w_k \mathcal{N}(m_k, \Sigma_{k,t}), \quad \Sigma_{k,t} \preceq \sigma_*^2 I, \quad \|m_k - m_\ell\| \geq \Delta \quad (k \neq \ell), \quad (6)$$

and for each k there exists some $\ell \neq k$ with $\|m_k - m_\ell\| \leq O(\Delta)$, and

$$p_t \left(\bigcup_{k=1}^{K_0} B(m_k, c\sigma_*) \right) \geq 1 - \delta_t$$

for some $c > 0$ and small $\delta_t \in [0, 1]$, where $B(m, r) = \{x \in \mathbb{R}^d : \|x - m\| \leq r\}$ denotes the Euclidean ball of radius r centered at m .

Under this definition, p_t is a mixture of narrow peaks $\{B(m_k, c\sigma_*)\}_{\forall k}$ separated by low-density gaps. For molecular data, the parameter σ_* is small, reflecting that each valid configuration is concentrated in a narrow neighborhood of configuration space. We provide a constructive chemical example and dataset-level scale comparison in Appendix C to support this characterization. At moderate timesteps t , forward noising smooths these peaks (see Eq. (2) and Eq. (3)) and creates overlap regions between them, thus a sample z_t may lie in the overlap, close to the midpoint between two peaks (see Fig. 1). Except at the exact symmetric midpoint, which has probability zero under the continuous intermediate distribution, a generic off-separator overlap state has unequal component responsibilities. In the regime derived in Appendix D, the score points toward a selected peak with magnitude $\|\nabla \log p(z_t)\| \sim \frac{\Delta}{\sigma_*^2}$ [Song et al., 2021b], and the reverse update step based on Eq. (5) is

$$\|z_{t-1} - z_t\|_{\text{det}} \approx \beta_t \cdot \frac{\Delta}{\sigma_*^2}. \quad (7)$$

Let m_k be the peak selected by the score direction, and define

$$u_k = \frac{m_k - z_t}{\|m_k - z_t\|}, \quad d_k(z_t) = \|m_k - z_t\|.$$

Under the approximation in Eq. (7), a sufficient condition for crossing the far boundary of the selected peak is

$$\beta_t |\langle \nabla \log p_t(z_t), u_k \rangle| > d_k(z_t) + c\sigma_* \implies z_{t-1} \notin B(m_k, c\sigma_*). \quad (8)$$

For a central off-separator overlap state, $d_k(z_t) = \Theta(\Delta)$, so this condition scales as $\beta_t \Delta / \sigma_*^2 > \Theta(\Delta) + c\sigma_*$. The derivation and toy examples are provided in Appendix D. In other words, when z_t originates from an overlap region created by forward noising, the reverse step is prone to push it across a thin peak and into a low-density region. Subsequent denoising cannot recover from this drift. For images, by contrast, peaks are broad (σ_* is large) and can overlap smoothly, so the condition in Eq. (8) is rarely triggered.

Consequently, the overshoot mechanism in Eq. (8), which arises directly from the concentration property in Definition 3.1, explains the fragility of reverse inference. The score field $\nabla \log p_t$ indeed points toward high-density peaks; however, because molecular peaks are narrow, the reverse update can step *past* the peak and cross the high-density into the opposite regions. **Once outside the distribution, subsequent updates are driven by the model score $\nabla \log q_t$ in a low-density region where estimation and discretization errors are large [Zhang et al., 2023, Li and van der Schaar, 2024], leading to oscillation or further drift rather than reliable re-entry into the correct peak.** Moreover, Cao et al. [2023]

also analyzed this re-entry problem and demonstrated the benefits of stochastic samplers, which

Table 1: Effect of starting timestep t on sample quality. $t = 0$ uses clean data; $t = 1000$ starts from pure Gaussian noise (standard diffusion). Intermediate t forms $z_t \sim p(z_t | x)$, and then we run t reverse steps for generated results. The experiment setting follows EDM on QM9. Higher numbers are better. Please refer to Sec. 4.1 for further details.

t	Atom Sta (%) $\uparrow$	Mol sta (%) $\uparrow$	Valid (%) $\uparrow$
0	99.0	95.2	97.7
100	99.0	92.7	96.4
300	98.9	89.1	95.5
500	98.7	86.2	94.3
1000	98.7	82.0	91.9

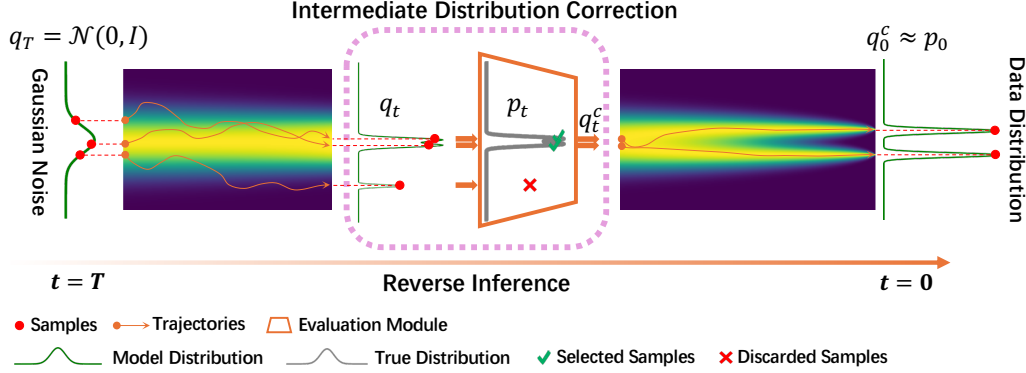


Figure 2: Illustration of **DIST**. In standard reverse inference, trajectories diffuse backward from Gaussian noise q_T toward the data distribution p_0 , but the model distribution q_t may drift away from the true distribution p_t due to the DC-structure of molecular data (see Sec. 3.1). At an intermediate timestep t , **DIST** steers q_t toward p_t by estimating consistent local regions and constructing a corrected marginal q_t^c . The resulting corrected distribution q_t^c better approximates p_t , realigning trajectories and improving both stability and validity in the final generation.

further underscores the importance of trajectory correction in SDE simulation. This phenomenon is more obvious in molecular generation due to the DC-structure, and we provide a detailed comparison and explanation specific to molecules in Appendix E.

In practice, discrepancies between the true marginal p_t and the model q_t accumulate across timesteps, and low-density region excursions become effectively unrecoverable. As shown in Table 1, inference quality degrades monotonically with t increasing, reflecting the growing deviation between p_t and q_t . This motivates the need for a corrective mechanism at intermediate timesteps to prevent off-distribution drift. An overview of our proposed method, **Diffuse and STeer (DIST)**, is illustrated in Fig. 2, and we formalize how DIST selectively realigns q_t with p_t in Sec. 3.2.

3.2 Diffuse and Steer

As discussed in Sec. 3.1 above, the unique characteristics of the molecular data distribution lead to severe inference and learning difficulties, such that the learned denoiser can be very inaccurate. As a result, the intermediate model distribution q_t often deviates from the true marginal distribution p_t . Moreover, during training, the diffusion model is trained on the true marginal distribution p_t from the dataset. In other words, the reverse process is implicitly learned under the assumption that the intermediate states follow the true marginals. Intuitively, when q_t drifts away from p_t , this will create a mismatch between the final distribution obtained by applying the reverse process to q_t and that obtained from p_t . Mathematically, we can show that this is true in Corollary 3.1 below.

Corollary 3.1 (TV-contraction Step). *Let $K_{t \rightarrow 0}$ be the ideal reverse Markov kernel, which can be intuitively understood as the perfect diffusion model with the true score functions; in other words, when the ideal reverse Markov kernel is applied to the true marginal distribution, we obtain the true data distribution $p_0 = K_{t \rightarrow 0}p_t$. Then, for the fixed pair (q_t, p_t) , there exists a pair-dependent TV-contraction coefficient $\kappa(q_t, p_t) \in [0, 1]$ such that*

$$\|q_0 - p_0\|_{\text{TV}} = \|K_{t \rightarrow 0}q_t - K_{t \rightarrow 0}p_t\|_{\text{TV}} \leq \kappa(q_t, p_t) \|q_t - p_t\|_{\text{TV}}. \quad (9)$$

where if q_t is the intermediate model distribution, q_0 can be understood as the final model distribution obtained by applying the perfect diffusion model on q_t .

The proof and explanation are deferred to the Appendix F.1. **Specifically, Corollary 3.1 reveals that if the intermediate model distribution q_t is closer to the true marginal distribution p_t , the final model distribution q_0 is closer to the true data distribution p_0 that we aim to obtain.** Therefore, to achieve high-quality generation despite the difficulties posed by the molecular data distribution, **our goal is to obtain an improved intermediate distribution q_t^c that remains closer to the true marginal p_t rather than blindly using the model distribution.** To achieve this

goal, we propose **DIST** (Diffuse and STeer), a corrective sampling approach for 3D molecular diffusion. Specifically, we perform the reverse process normally as in the standard diffusion pipelines; however, we incorporate an additional correction step to steer the intermediate distribution q_t toward a “corrected” version q_t^c closer to the true marginal p_t . An overview of DIST is provided in Fig. 2.

We now present the details of DIST concretely. Building on Definition 3.1, which states that the distribution p_t concentrates around a finite number of peaks separated by low-density regions, we next introduce a finer partition of the support into small neighborhoods. Specifically, we divide the space into radius- r batches $\{B_j\}_{j=1}^J$, which can be regarded as local regions within or around the peaks, each carrying probability mass

$$\pi_j := p_t(B_j), \quad \hat{\pi}_j := q_t(B_j),$$

together with the conditional distributions $p_{t|j}$ and $q_{t|j}$ restricted to each batch B_j .

Each batch j is further associated with a model-side pilot score $s_j \in \mathbb{R}$ (e.g., round-trip residual, self-consistency, ensemble variance, or chemistry-based penalty), which reflects whether the region is consistent with the true marginal distribution or potentially invalid. Given a threshold τ , we select batches whose scores fall above τ :

$$J^*(\tau) := \{j : s_j \geq \tau\}.$$

We then measure how much probability mass remains after this selection by defining

$$\alpha(\tau) := \sum_{j \in J^*(\tau)} \pi_j, \quad \beta(\tau) := \sum_{j \in J^*(\tau)} \hat{\pi}_j.$$

Here, $\alpha(\tau)$ represents the *true coverage*, i.e., the portion of the ground-truth distribution preserved by the selection, while $\beta(\tau)$ denotes the *model coverage*, i.e., the portion of the model distribution retained. Larger thresholds τ restrict the selection to batches that are more likely to correspond to valid regions; smaller thresholds broaden the selection and capture more mass, but at the cost of admitting regions inconsistent with the true distribution. The selected model distribution at threshold τ is then given by

$$q_t^c(\tau) := \sum_{j \in J^*(\tau)} \tilde{\pi}_j q_{t|j}, \quad \tilde{\pi}_j = \frac{\hat{\pi}_j}{\sum_{k \in J^*(\tau)} \hat{\pi}_k}. \quad (10)$$

Intuitively, q_t consists of both samples consistent with p_t , lying within valid regions, and samples that fall outside. The corrected distribution q_t^c is a reweighted intermediate marginal that improves approximation to p_t before the remaining reverse kernel is applied. The following proposition establishes a quantitative error bound that illustrates the effectiveness of DIST.

Proposition 3.1 (Selective Reverse Error Bound). *Under the DC-structure in Definition 3.1 and the batch construction described above, for any threshold τ the deviation between the selectively corrected reverse distribution $K_{t \rightarrow 0} q_t^c(\tau)$ and the true distribution $p = K_{t \rightarrow 0} p_t$ admits an upper bound of the form*

$$\|K_{t \rightarrow 0} q_t^c(\tau) - p\|_{\text{TV}} \leq f(\alpha(\tau), \beta(\tau), (\pi_j, \hat{\pi}_j)_{j \in J^*(\tau)}, \sup_{j \in J^*(\tau)} \text{TV}(q_{t|j}, p_{t|j})), \quad (11)$$

where $f(\cdot)$ is an explicit function of the true coverage $\alpha(\tau)$, the model coverage $\beta(\tau)$, the selected batch weights, and the conditional discrepancies.

The exact form of $f(\cdot)$ and the proof are provided in Appendix F.2. This error bound provides a theoretical guarantee for DIST; that is, **selective correction ensures that q_t^c is steered toward convergence with the true distribution p at intermediate timestep t , stabilizing the sampling trajectory.**

Corrective Sampling. We now describe how the corrected distribution q_t^c is achieved in the reverse inference procedure (see Fig. 2). At a given intermediate timestep t , DIST constructs a candidate pool by reverse-simulating a small set of samples from Gaussian noise at T . Each candidate is duplicated and perturbed with a sufficiently small amount of noise to form batches $\{B_j\}_{j=1}^J$, which collectively follow the model distribution q_t and remain within the prescribed radius- r constraint (see Definition 3.1). To evaluate whether these batches $\{B_j\}_{j=1}^J$ are consistent with the true distribution

p_t , DIST runs a full reverse inference on a pilot subset $\{B_j^{\text{sub}} \mid B_j^{\text{sub}} \subset B_j\}_{j=1}^J$ drawn from each batch. This pilot inference provides an empirical assessment of how well the current model trajectory aligns with p_t , and serves as a diagnostic of potential drift away from the true distribution. Based on the pilot outcomes $s_j \in \mathbb{R}$, DIST reweights local batches by $\hat{\pi}_j$ under a universal threshold τ , obtaining a corrected distribution $q_t^c(\tau)$ (see Eq. (10)) that better approximates p_t . In effect, q_t^c concentrates the reverse trajectories around valid molecular peaks. Beyond improved approximation quality, DIST also provides an efficiency advantage by reducing unnecessary inference on invalid regions, as demonstrated in Sec. 4.3. The pseudocode for DIST is provided in Algorithm 1.

Since DIST duplicates and perturbs intermediate candidates, a natural concern is whether the resulting samples become overly similar. However, DIST keeps valid high-coverage regions under the DC-structure, rather than forcing samples into a single mode. Empirically, the uniqueness metric in Sec. 4 shows that **DIST preserves diversity while improving stability and validity**.

4 Experiments

4.1 Setups

Datasets. Following prior work [Hoogetboom et al., 2022, Xu et al., 2023, Song et al., 2024], we evaluate DIST on two widely used standard datasets in molecular generation: QM9 [Ramakrishnan et al., 2014] and GEOM-Drugs [Axelrod and Gómez-Bombarelli, 2022]. QM9 contains 130K small molecules, restricted to at most 9 heavy atoms (29 atoms including hydrogen atoms). We follow the standard partition from Hoogetboom et al. [2022], with 100K molecules for training, 18K for validation, and 13K for testing. GEOM-Drugs is substantially larger, comprising 420K molecules with an average of 44.4 atoms and up to 181 atoms. We retain the 30 lowest-energy conformations for each molecule.

Metrics. Strictly following prior work, we evaluate generated molecules using the standard metrics: atom stability, molecule stability, validity, and validity $\times$ uniqueness² [Simonovsky and Komodakis, 2018, Garcia Satorras et al., 2021]. *Atom Stability*: the percentage of atoms whose number of bonds matches their valence (e.g., H:1, C:4, O:2). *Molecule Stability*: the percentage of molecules in which all atoms are stable. *Validity*: the percentage of molecules satisfying valence rules for all atoms. *Uniqueness*: the percentage of molecules that are distinct from one another, serving as **the standard diversity metric in 3D molecular generation**. Note for GEOM-Drugs, following prior work, we omit the stability and uniqueness metrics, since they are consistently close to 0% and 100%, respectively, for all evaluated methods including the baseline methods.

Baselines and Implementation Details. We employ several representative state-of-the-art diffusion models for 3D molecular generation, including EDM [Hoogetboom et al., 2022], GeoLDM [Xu et al., 2023], and RADM_{DIT-B} [Ding and Hofmann, 2025], as backbone models to evaluate the benefits from the employment of DIST and compare with the original without DIST. These backbones span both equivariant/non-equivariant, GNN/Transformer, and latent/non-latent space approaches, **ensuring DIST is evaluated across diverse architectures**. In addition, we include comparisons with well-known non-diffusion-based models, such as ENF [Garcia Satorras et al., 2021] and G-SchNet [Gebauer et al., 2019]. The results of backbone models and baseline methods are directly obtained from their original work. Specifically, we strictly use the officially released model weights without altering any hyperparameters or settings. For fair comparison, we also exclude the methods that incorporate chemical priors (e.g., Open Babel) in postprocessing. For detailed experimental settings, please refer to Appendix G.

4.2 Main Results and Analysis

DIST corrects the intermediate model distribution toward the true marginal p_t . Since p_t is intractable in high-dimensional molecular generation, directly measuring the correction is infeasible. We therefore demonstrate its effectiveness through final generation quality, as supported by our theory. Following prior work, we use the QM9 and GEOM-Drugs datasets, and generate 10,000 3D molecules from each model. The main results are summarized in Table 2. Consistent with prior work, for

²Reporting uniqueness together with validity avoids overestimating diversity when many generated samples are invalid.

Table 2: Results for atom stability, molecule stability, validity, and validity $\times$ uniqueness. Higher values indicate better performance. Check Sec. 4.1 for experimental setup details. All models combined with DIST surpass their original counterparts, and the improved results are shown in **bold**. Global best results are underlined.

Metrics	QM9				GEOM-Drugs	
	Atom Sta (%)	Mol Sta (%)	Valid (%)	Valid $\times$ Unique (%)	Atom Sta (%)	Valid (%)
Data	99.0	95.2	97.7	97.7	86.5	99.9
ENF	85.0	4.9	40.2	39.4	-	-
G-SchNet	95.7	68.1	85.5	80.3	-	-
EDM	98.7	82.0	91.9	90.7	81.3	92.6
EDM+DIST	99.2$\pm$0.0	89.9$\pm$0.3	96.9$\pm$0.2	94.1$\pm$0.3	82.2	96.0
GeoLDM	98.9	89.4	93.8	92.7	84.4	99.3
GeoLDM+DIST	99.4$\pm$0.0	93.4$\pm$0.3	96.3$\pm$0.2	93.1$\pm$0.2	85.4	99.7
RADM	98.5	87.3	94.1	91.7	85.0	99.3
RADM+DIST	99.1$\pm$0.0	91.4$\pm$0.3	96.2$\pm$0.1	92.3$\pm$0.4	86.0	99.8

QM9, we report averages over three runs together with standard deviations; for GEOM-Drugs, the deviations are negligible and thus omitted. Across both datasets and all metrics, every backbone model combined with DIST consistently outperforms its original counterpart. The improvements are significant and universal: **all bold numbers in Table 2 indicate that DIST significantly improves the quality of generated molecules, with particularly large margins observed on the most critical stability metrics. The improved validity $\times$ uniqueness further shows that DIST increases the number of valid and diverse generated molecules.** In addition, methods based on our DIST set the new state-of-the-art for molecular diffusion generation on both QM9 and GEOM-Drugs dataset. Notably, the margins of improvement observed before and after applying our method highlight the generality of DC-structure issue. Across GNN-based equivariant EDM [Hoogeboom et al., 2022], GeoLDM [Xu et al., 2023] and Transformer-based non-equivariant RADM [Ding and Hofmann, 2025], where GeoLDM and RADM perform in latent space, the issue remains consistently evident. This observation cautions against relying solely on architectural choices. Our experimental results confirm that, as a plug-in component, DIST effectively steers inference trajectories and thus mitigates distributional discrepancies in the sampling process, providing a valuable complement to architectural innovations to improve generation quality.

4.3 Efficiency Analysis

Since the batches $\{B_j\}_{j=1}^J$ are created by duplication and perturbation, DIST requires only $\frac{T-t}{|B|}$ expected timesteps per inference from T (1000 is adopted in backbone models) to t , where $|B|$ is the batch size. For example, setting $t = 300$ with $|B| = 100$, each selected batch after threshold correction requires only $307 (\frac{1000-300}{100} + 300)$ steps instead of the 1000 steps as used in standard counterparts. A detailed comparison of efficiency is provided in Table 3, which shows DIST can substantially reduce the overall timestep by nearly **half** compared to baselines, while **significantly improving the generation quality** as shown in Table 2. We also provide a detailed quantification of the expected computational cost of our DIST in Appendix H.1 and wall-clock time in Appendix H.2.

Table 3: Average number of timesteps required for a full inference procedure. The values are computed from the total timestep consumption needed to generate 10,000 molecules, corresponding to the experiments in Table 2. All baseline methods use the standard 1000-step schedule, whereas DIST significantly reduces the computational cost.

Methods	QM9	GEOM-Drugs
EDM+DIST	556.1	503.3
GeoLDM+DIST	416.9	636.7
RADM+DIST	413.7	438.8
Baselines	1000	1000

4.4 Ablation Study

The number of pilot samples drawn from each batch plays a critical role. A larger set of pilot samples provides a more accurate representation of the model distribution q_t , and leads to a better corrected distribution q_t^c by DIST. However, increasing the number of pilot samples also leads to higher computational costs. In practice, we may choose a pilot set size that is sufficiently representative while

remaining computationally affordable. We conduct an ablation study to compare the final sample quality and computational costs under different numbers of pilot samples, with results reported in Table 4.

As expected, increasing the number of pilot samples improves the quality of generated molecules monotonically. At the same time, computational costs (measured by the number of time steps) also increase monotonically. Nevertheless, even under a relatively small budget (30, 50, 100), DIST still demonstrates superior performance, significantly improving the baseline in both sample quality and computational efficiency. Moreover, we also conducted ablation study on all hyperparameters, including batch score threshold, intermediate timestep, and perturbation intensity, as shown in Appendix I.

Table 4: Ablation study for varying pilot subset sizes using EDM+DIST on QM9 with a fixed batch size of 100. We report the generation performance and the average number of timesteps.

Size	Atom Sta (%)	Mol Sta (%)	Valid (%)	Valid×Unique (%)	Timesteps
30	99.2	89.5	96.7	94.3	428.3
50	99.2	89.9	96.9	94.1	556.1
100	99.3	90.5	97.3	94.9	644.7

5 Conclusion and Future Work

In this work, we investigated the unique challenge of applying diffusion models to molecular generation. Molecular data are confined to concentrated regions of the representation space, with chemically valid structures corresponding to densely packed sharp peaks separated by regions of near-zero density. This DC-structure makes diffusion modeling fragile, since small errors at intermediate timesteps are amplified, causing generative trajectories to drift off-distribution and accumulate irreparable structural violations. To address this issue, we proposed DIST, which selectively reweights intermediate model marginals and steers inference trajectories toward valid molecular peaks. DIST is model-agnostic and can be integrated into a wide range of diffusion-based molecular generators. We also provided both theoretical analysis and experimental results to demonstrate that our method consistently improves the performance across multiple architectures for molecular generation, while nearly halving the inference cost.

Limitations and future directions First, although DIST is a general corrective framework, its validation is limited to 3D molecular generation; extension to other domains requires verifying whether similar DC-structure exists and whether reliable diagnostics are available. Protein generation is a more challenging and fundamentally different task, involving larger structures, hierarchical organization, long-range interactions, and more complex validity constraints. Second, while our study focuses on diffusion models, DC-structure may also appear in other generative paradigms. Exploring corrective strategies in normalizing flows [Rezende and Mohamed, 2015], autoregressive models [Li et al., 2024b], or energy-based frameworks [Du and Mordatch, 2019] can provide a broader principle for modeling constrained distributions.

Acknowledgments

This work was supported in part by the U.S. National Science Foundation under Award No. 2551109.

GenAI Usage Statement

LLMs are used to assist with grammar checking, improving writing fluency, managing computational resources and providing suggestions after the initial draft. All observations, ideas, methodologies in this paper are developed entirely by the authors. Any content generated with the help of LLMs is created under detailed author instructions and thoroughly verified by the authors before inclusion.

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A Molecular Diffusion Models

Molecular diffusion models must account for the fact that 3D molecular structures are naturally defined up to rigid-body transformations. While atomic features are scalar quantities invariant to translations and rotations (i.e., $\mathbb{SE}(3)$ -transformations), the atomic coordinates transform equivariantly under these operations [Thomas et al., 2018, Hooeboom et al., 2022, Dumitrescu et al., 2025].

However, arbitrary $\mathbb{SE}(3)$ transformations can pose challenges for standard denoising networks, since the same molecule under a different rotation or translation may be treated as a distinct input. To overcome this issue, existing works often incorporate $\mathbb{SE}(3)$ -equivariant frameworks to ensure symmetry-awareness. In particular, translational invariance can be enforced by subtracting the centroid of the atomic coordinates, thereby removing global translational degrees of freedom [Garcia Satorras et al., 2021, Xu et al., 2022]. Handling rotations is more subtle, and is typically achieved either through carefully designed equivariant neural architectures [Hooeboom et al., 2022, Xu et al., 2023] or through canonicalization procedures that align molecules to a reference orientation [Ding and Hofmann, 2025, Kaba et al., 2023, Rempe et al., 2020].

In addition, the hybrid discrete–continuous nature of molecular data [Dunn and Koes, 2024] introduces further challenges for generative modeling. Several recent works address this difficulty by learning smoother latent representations [Xu et al., 2023, Ding and Hofmann, 2025, Chen et al., 2025, Luo et al., 2025]. These approaches commonly adopt a VAE-based encoder–decoder framework [Kingma and Welling, 2014], performing the diffusion process in a latent space rather than directly over atomic coordinates and features. While this can alleviate some modeling challenges, latent diffusion methods may introduce additional approximation error, and discrepancies can remain between generated samples and chemically valid molecular structures.

B Comparison with Corrective Diffusion Sampling and Rejection/Filtering Methods

In this section, we compare DIST with two related but distinct lines of work: corrective diffusion sampling and general rejection/filtering methods. The former studies how to improve diffusion inference or reduce sampling-time errors, while the latter provides broad sampling principles for accepting, rejecting, or updating candidates.

B.1 Comparison with Corrective Diffusion Sampling

One innovation of DIST lies in modeling the molecular distribution within the diffusion process as a DC-structure (Definition 3.1). This idea is conceptually related to the notion of a supervision region introduced in the recent work [Song et al., 2025]. However, that work interprets the phenomenon mainly as a factor affecting the generalization ability of diffusion trajectories, without addressing its impact on constrained and complex data types such as molecular structures, nor proposing any corrective mechanism. In contrast, DIST explicitly quantifies and mitigates this issue through a theoretically grounded correction process.

To the best of our knowledge, there is currently **no existing corrective method that directly steers intermediate distributions in diffusion-based molecular generation**. Two recent studies in the text-to-image domain, Dynamic CFG [Papalampidi et al., 2025] and ELECT [Kim et al., 2025], adopt selective sampling strategies that are loosely related to our idea. Dynamic CFG performs a greedy search to select the scale that maximizes evaluation scores at each step, while ELECT selects the best candidate from a pool at an intermediate timestep and denoises it as the final output. However, both works focus on sample-level quality refinement rather than distributional correction, lack theoretical grounding, and incur additional computational overhead due to repeated candidate discarding.

A separate line of recent work investigates exposure-bias effects in diffusion models [Ning et al., 2024, Wang et al., 2025, Li et al., 2024a]. These methods primarily analyze the general training-inference mismatch issue that arises during denoising: they study how prediction errors accumulate across reverse steps and how to make individual transitions more stable. Such analyses focus on *local transition dynamics*, such as characterizing variance inflation from prediction [Ning et al., 2024], identifying consistent neighborhoods of training samples [Wang et al., 2025], or aligning adjacent timesteps to reduce mismatch [Li et al., 2024a]. In contrast, DIST is motivated from a *distribution-level* perspective. Rather than attributing instability to step-wise prediction errors

alone, DIST formalizes the dense-concentrated structure of molecular data and identifies it as a fundamental source of fragility in molecular diffusion. The theoretical results of DIST therefore target *global distributional correctness*: by correcting the intermediate model distribution, DIST provides guarantees on the quality of the final distribution at $t = 0$.

Methodologically, existing corrective diffusion methods usually require modified training objectives, adversarial components, adjusted schedules, guidance-scale search, or candidate selection designed for images. DIST instead introduces a *training-free, plug-in selective correction module* that can be applied directly to existing molecular diffusion backbones and achieves NFE-level efficiency gains without altering their training objectives. Therefore, DIST is complementary to prior corrective diffusion methods but differs in its target: it corrects the intermediate distribution q_t toward the true marginal p_t , rather than only refining samples or stabilizing local denoising transitions.

B.2 Comparison with Rejection and Filtering Methods

DIST is also related to rejection and filtering at the technical-workflow level because it evaluates candidate regions and removes invalid ones. However, rejection/filtering is a much broader concept than DIST. Classical rejection sampling and sequential Monte Carlo methods provide general sampling principles for accepting, rejecting, or updating samples [Von Neumann, 1951, Gordon et al., 1993]. Related selection or resampling ideas also appear in modern generative systems, including EAGLE for speculative language-model decoding, GLIDE for guided diffusion generation, and RePaint for diffusion inpainting [Li et al., 2024c, Nichol et al., 2022, Lugmayr et al., 2022]. These examples show that selection-like operations are common building blocks in generative modeling, but they do not by themselves define a complete method for molecular diffusion.

Directly applying generic rejection/filtering to our setting has two limitations. First, standard rejection is usually final-sample based: it generates complete samples and then accepts or rejects them. This does not correct the intermediate model distribution q_t , where molecular diffusion trajectories may already have drifted away from the true marginal p_t . For DC-structured molecular data, such drift is especially harmful because valid regions are narrow and low-density invalid regions are difficult to recover from in later denoising steps. Second, generic rejection or resampling can incur substantial extra cost, since rejected candidates may require full sampling attempts.

DIST uses a filtering-like operation for a different purpose. Instead of treating filtering as post-hoc quality control, DIST uses it to construct a corrected intermediate distribution q_t^c before the remaining reverse process. It also shares the reverse computation from T to t , forms local batches at the intermediate timestep, and only continues selected regions from t to 0. Thus, DIST is not ordinary rejection sampling. It is a domain-specific, theory-grounded, and efficient correction mechanism designed for DC-structured molecular diffusion.

In summary, being the first training-free plug-in corrective module tailored for 3D molecular diffusion, DIST differs from both lines of related work in three aspects: (1) it identifies DC-structure as a molecular-specific source of diffusion fragility; (2) it corrects the intermediate distribution q_t with a solid theoretical foundation rather than only final samples or local denoising transitions; and (3) it incorporates an explicitly designed efficiency strategy that yields consistent improvements in both generative performance and computational cost over existing baselines.

C Chemical Evidence for DC-Structure

This section provides constructive chemical evidence for the DC-structure in Definition 3.1. The key observation is that, for any molecule in QM9 and GEOM-Drugs, a nearby yet distinct valid molecule can be systematically constructed through minimal local chemical changes. This supports the view that valid molecular configurations are not only highly concentrated, but also densely located in representation space.

One concrete example is **fluorine substitution** [Shimizu and Hiyama, 2005]. For any molecule containing a C–H bond, which is broadly applicable due to valency rules [Cartmell and Fowles, 2013] and the prevalence of C–H bonds in organic molecules (QM9 and GEOM-Drugs), we can replace the hydrogen atom with fluorine to obtain a nearby yet distinct valid molecule. The bond lengths are 1.09 Å for C–H and 1.35 Å for C–F [Ramakrishnan et al., 2014, Hoogeboom et al., 2022]. Under a conservative stability margin of 0.1 Å [Landrum, 2013, Hoogeboom et al., 2022], this substitution

Table 5: Scale comparison for a local valid chemical modification. The bond-change value corresponds to the residual perturbation induced by fluorine substitution.

Dataset	Mean Diameter (Å)	Avg. # Atoms	Bond Change (Å)	Relative Change
QM9	6.54	18.0	0.06	0.051%
GEOM-Drugs	12.32	44.4	0.06	0.011%

requires only

$$1.35 - 1.09 - 0.2 = 0.06 \text{ Å} \quad (12)$$

of residual coordinate perturbation while preserving chemical validity [Cartmell and Fowles, 2013]. Therefore, this local modification gives a concrete example where a distinct valid molecule can be reached with only a $\sim 0.06 \text{ Å}$ perturbation. In the notation of Definition 3.1, this corresponds to a small inter-peak change, i.e., $0.06 \text{ Å} \sim O(\Delta)$, directly supporting the condition that nearby valid molecular peaks can satisfy $\|m_k - m_\ell\| \leq O(\Delta)$.

We further compare this perturbation with the molecular scale. We define the molecular diameter as the maximum atom-to-atom distance within a molecule and report its dataset mean. As shown in Table 5, the required bond-level change is negligible relative to the molecular scale: far below 0.1% for both QM9 and GEOM-Drugs.

This perturbation is negligible at the molecular scale, yet it produces a distinct valid molecule. This demonstrates that valid configurations can be highly concentrated and densely arranged, as assumed in the DC-structure. Such behavior is fundamentally different from non-DC distributions such as images, where changing semantic meaning typically requires much larger perturbations.

D Overshoot Mechanism

In this section, we provide the derivation of overshoot mechanism in Appendix D.1 and toy examples for it in Appendix D.2.

D.1 Update Issue

We justify the scaling of score magnitude in the overlap region and reverse update length in Eq. (7) and Eq. (8) of the main text. Under the DC-structure (see Definition 3.1),

$$p_t \simeq \sum_{k=1}^{K_0} w_k \mathcal{N}(m_k, \Sigma_{k,t}), \quad \Sigma_{k,t} \preceq \sigma_*^2 I, \quad \|m_k - m_\ell\| \geq \Delta \quad (k \neq \ell),$$

and define the responsibilities

$$\gamma_k(z_t) = \frac{w_k \mathcal{N}(z_t; m_k, \sigma_*^2 I)}{\sum_j w_j \mathcal{N}(z_t; m_j, \sigma_*^2 I)}.$$

Then

$$\nabla \log p(z_t) = \frac{1}{\sigma_*^2} \left(\sum_k \gamma_k(z_t) m_k - z_t \right).$$

To account for the symmetric case, consider two equally weighted isotropic components centered at $m_\pm = \bar{m} \pm \frac{\Delta}{2} u$, and write $z_t = \bar{m} + yu$. The exact score along the inter-peak direction is

$$\langle \nabla \log p_t(z_t), u \rangle = \frac{1}{\sigma_*^2} \left[\frac{\Delta}{2} \tanh \left(\frac{\Delta y}{2\sigma_*^2} \right) - y \right].$$

This score is zero at the exact midpoint $y = 0$. However, at the natural responsibility-transition scale $|y| = \Theta(\sigma_*^2/\Delta)$, its magnitude is already $\Theta(\Delta/\sigma_*^2)$. Hence, for generic off-separator overlap states,

$$\|\nabla \log p_t(z_t)\| = \Theta \left(\frac{\Delta}{\sigma_*^2} \right), \quad (13)$$

which is the scaling used in the main text. And based on the reverse update step used in main text (see equation 5), the deterministic displacement satisfies

$$\|z_{t-1} - z_t\|_{\text{det}} \approx \beta_t \|\nabla \log p(z_t)\|.$$

The detailed derivation is provided in Appendix D.3. Combining this scaling with the deterministic displacement gives the full sufficient condition in Eq. (8), which includes both the distance to the selected center and the peak radius.

D.2 Toy Examples

To further illustrate the effect of DC-structure (see Definition 3.1) on diffusion sampling, we provide two controlled toy experiments based on Mixture-of-Gaussians (MoG): one in 2D and one in 1D. In each case, we construct *two distributions under identical diffusion settings* (same noise schedule, network architecture, optimizer, and sampling steps): (i) a smooth MoG with well-separated modes, and (ii) a DC-structured MoG with narrow and closely packed peaks. Brighter colors in the heatmaps correspond to higher density regions.

2D MoG: Effect of DC-structure. The DC-structured MoG (‘Narrow multi-peak MoG’ in Fig. 3) contains sharply concentrated peaks placed in close proximity, mimicking the clustered geometric modes observed in molecular data. As shown in Fig. 3, under identical diffusion parameters, sampling from this DC-structured distribution exhibits noticeably poorer behavior: a substantial portion of generated samples drift into low-density regions. This demonstrates that DC-structure imposes additional instability on the reverse diffusion process, even in simple synthetic settings.

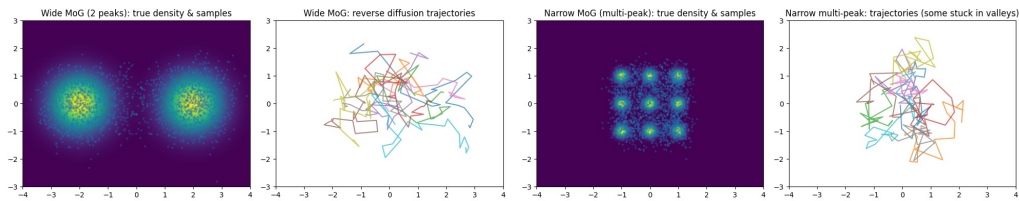


Figure 3: 2D Mixture-of-Gaussians examples. Left: smooth MoG. Right: DC-structured MoG.

1D MoG: Overshoot phenomenon. The 1D DC-structured MoG (‘Sharp 4-peak mixture’ in Fig. 4) contains four narrow peaks in close proximity. As illustrated in Fig. 4, at intermediate timesteps (around 25%–50% of the reverse process before generated results), some trajectories *overshoot*: they cross peak regions and fall into low-density regions, ultimately producing invalid samples. This behavior aligns with the overshoot mechanism analyzed in Sec. 3.1.

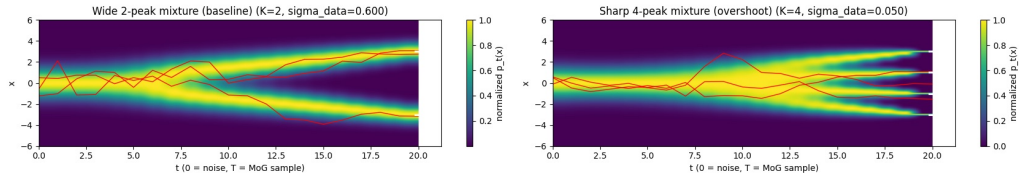


Figure 4: 1D Mixture-of-Gaussians examples. Left: smooth MoG. Right: DC-structured MoG exhibiting overshoot.

These synthetic experiments empirically support our theoretical analysis: *DC-structured distributions are significantly more fragile under diffusion sampling*, consistently exhibiting drift, overshoot, and degraded sample quality. This behavior directly parallels challenges found in molecular diffusion models and further motivates the formulation of DIST as a corrective sampling framework for DC-structured data.

D.3 Deterministic Displacement

In order to analyze the overshoot behavior induced by the DC-structure of $p(z_t)$, it is necessary to isolate the *deterministic displacement* of the reverse update in Eq. (5). The stochastic noise term is

not included in this analysis, because an additive Gaussian perturbation can move a trajectory in an arbitrary direction and therefore obscures the geometric effect we seek to characterize. Our interest is in the model-induced geometric drift of the reverse process, not in the random diffusive fluctuations. For this reason, we study the deterministic quantity $\|z_{t-1} - z_t\|_{\text{det}}$, which captures the intrinsic drift responsible for overshoot, and we provide a detailed derivation below.

Isolate the deterministic displacement of the reverse DDPM update used in equation 5:

$$z_{t-1} = \frac{1}{\sqrt{1-\beta_t}} \left(z_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(z_t, t) \right).$$

Subtracting z_t and regrouping terms gives

$$\begin{aligned} z_{t-1} - z_t &= \frac{1}{\sqrt{1-\beta_t}} z_t - \frac{\beta_t}{\sqrt{1-\beta_t}\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(z_t, t) - z_t \\ &= \underbrace{\left(\frac{1}{\sqrt{1-\beta_t}} - 1 \right)}_{A_t} z_t - \underbrace{\frac{\beta_t}{\sqrt{1-\beta_t}\sqrt{1-\bar{\alpha}_t}}}_{B_t} \varepsilon_\theta(z_t, t). \end{aligned}$$

Thus the coefficients in front of z_t and the score term are, respectively,

$$A_t = \frac{1}{\sqrt{1-\beta_t}} - 1 = \frac{1 - \sqrt{1-\beta_t}}{\sqrt{1-\beta_t}}, \quad B_t = \frac{\beta_t}{\sqrt{1-\beta_t}\sqrt{1-\bar{\alpha}_t}}.$$

For example, EDM [Hoogeboom et al., 2022] and GeoLDM [Xu et al., 2023] use a polynomial noise schedule, under which $\beta_{300} \approx 1.6 \times 10^{-3}$. This confirms that β_t remains very small at moderate timesteps.

Relative scale of z_t and ε_θ . During training, under the forward process (see Eq. (4)), both z_0 and ε have $O(1)$ variance. Thus for typical samples

$$\|z_t\| \asymp \|\varepsilon\| \asymp \|\varepsilon_\theta(z_t, t)\|,$$

since the DDPM noise-prediction objective ensures that $\varepsilon_\theta(z_t, t)$ matches the distribution of ε . Hence

$$\|A_t z_t\| \asymp |A_t|, \quad \|B_t \varepsilon_\theta(z_t, t)\| \asymp |B_t|.$$

Asymptotic dominance of the score-driven transport term. A Taylor expansion gives

$$A_t = \frac{\beta_t}{2} + O(\beta_t^2), \quad \frac{1}{\sqrt{1-\beta_t}} = 1 + O(\beta_t),$$

so

$$B_t = \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} (1 + O(\beta_t)) \asymp \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}.$$

Therefore

$$\frac{\|A_t z_t\|}{\|B_t \varepsilon_\theta(z_t, t)\|} \asymp \frac{|A_t|}{|B_t|} = \frac{\frac{\beta_t}{2} + O(\beta_t^2)}{\beta_t / \sqrt{1-\bar{\alpha}_t}} = \frac{1}{2} \sqrt{1-\bar{\alpha}_t} + O(\beta_t).$$

Since $1 - \bar{\alpha}_t \rightarrow 0$ as $t \rightarrow 0$, we obtain the limit

$$\lim_{t \rightarrow 0} \frac{\|A_t z_t\|}{\|B_t \varepsilon_\theta(z_t, t)\|} = 0,$$

i.e. the Gaussian contraction drift is asymptotically negligible compared to the score-driven transport term at a moderate t .

Final approximation. Using the identity

$$\nabla \log p(z_t) = -\frac{1}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(z_t, t),$$

the $1/\sqrt{1-\bar{\alpha}_t}$ factor is absorbed into the score parameterization, so that $B_t \varepsilon_\theta$ becomes a coefficient of order β_t in front of $\nabla \log p(z_t)$. Thus the deterministic displacement satisfies

$$\|z_{t-1} - z_t\|_{\text{det}} \approx \beta_t \|\nabla \log p(z_t)\|,$$

with the Gaussian contraction drift $A_t z_t$ being strictly lower order in the small-step regime.

E Supplementary Experiments

The experiment shown in Table 6 is designed to illustrate how stochastic correction in diffusion sampling interacts with the DC-structure of molecular data. In molecular distributions characterized by densely packed and sharply concentrated peaks (Sec. 3.1), even small prediction errors can cause sampling trajectories to drift into low-density regions. SDE-based samplers such as DDPM [Ho et al., 2020] introduce random perturbations that help trajectories re-enter valid high-density regions, effectively compensating for such errors [Cao et al., 2023]. In contrast, ODE-based samplers like DDIM [Song et al., 2020] are fully deterministic, lacking this corrective mechanism and thus being more susceptible to error accumulation when the distribution is highly concentrated.

To test this hypothesis, we compare the generation behavior of molecular data (with DC-structure) and image data (with smoother distributions) under both DDIM and DDPM sampling. As shown in Table 6, **DDIM consistently outperforms DDPM in image generation (CIFAR10), whereas the opposite trend appears for molecular generation (QM9), where DDPM yields substantially higher molecule stability across all timestep settings**. These results indicate that the stochasticity in DDPM sampling mitigates overshoot and helps realign trajectories with valid molecular peaks. This finding supports our theoretical analysis in Sec. 3.1, which identified DC-structure as a key factor causing fragility in reverse inference, and it further motivates the design of our proposed DIST method, which explicitly corrects intermediate distributions to achieve similar stabilization effects in a principled manner.

Table 6: Comparison between CIFAR10 image generation (measured by FID) and QM9 molecular generation (measured by molecule stability) under DDIM [Song et al., 2020] and DDPM [Ho et al., 2020] sampling. Results are reported for different numbers of timesteps S . Lower is better for FID, higher is better for stability. CIFAR10 numbers are taken from the original DDIM paper [Song et al., 2020], and QM9 experiments follow the setup of EDM [Hoogeboom et al., 2022]. The better values are shown in **bold**.

S	CIFAR10 FID ↓					QM9 Molecule Stability (%) ↑				
	10	20	50	100	1000	10	20	50	100	1000
DDIM	13.36	6.84	4.67	4.16	4.04	9.6	37.4	57.0	63.0	65.9
DDPM	41.07	18.36	8.01	5.78	4.73	12.4	50.9	74.1	77.9	82.0

F Proofs

Here we provide the proofs of Corollary 3.1 and Proposition 3.1.

F.1 Corollary 3.1

We first give the preliminary knowledge about the following corollary then derive the proof.

Total Variation (TV) distance. For two probability measures μ, ν on a measurable space $(X, \mathcal{F})$, the *total variation distance* is defined by

$$\|\mu - \nu\|_{\text{TV}} := \sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)|.$$

Equivalently, by the Hahn–Jordan decomposition, it admits the dual characterization

$$\|\mu - \nu\|_{\text{TV}} = \frac{1}{2} \sup_{\|f\|_{\infty} \leq 1} \left| \int_X f d\mu - \int_X f d\nu \right|,$$

where the supremum is over all measurable test functions f with $\|f\|_{\infty} \leq 1$.

Contraction under Markov kernels. Let K be a Markov kernel, i.e. $K(x, \cdot)$ is a probability measure for each $x \in X$, and $x \mapsto K(x, A)$ is measurable for each $A \in \mathcal{F}$. For a probability measure μ , define the pushforward

$$(K\mu)(A) := \int_X K(x, A) \mu(dx), \quad A \in \mathcal{F}.$$

For a bounded measurable function $f : X \rightarrow \mathbb{R}$, define

$$(Kf)(x) := \int_X f(y) K(x, dy).$$

Note that $|Kf(x)| \leq \|f\|_\infty$, hence $\|Kf\|_\infty \leq \|f\|_\infty$.

Corollary 3.1 (TV-contraction Step). *Let $K_{t \rightarrow 0}$ be the ideal reverse Markov kernel, which can be intuitively understood as the perfect diffusion model with the true score functions; in other words, when the ideal reverse Markov kernel is applied to the true marginal distribution, we obtain the true data distribution $p_0 = K_{t \rightarrow 0} p_t$. Then, for the fixed pair (q_t, p_t) , there exists a pair-dependent TV-contraction coefficient $\kappa(q_t, p_t) \in [0, 1]$ such that*

$$\|q_0 - p_0\|_{\text{TV}} = \|K_{t \rightarrow 0} q_t - K_{t \rightarrow 0} p_t\|_{\text{TV}} \leq \kappa(q_t, p_t) \|q_t - p_t\|_{\text{TV}}. \quad (9)$$

where if q_t is the intermediate model distribution, q_0 can be understood as the final model distribution obtained by applying the perfect diffusion model on q_t .

Proof. For any two probability measures μ, ν ,

$$\|K\mu - K\nu\|_{\text{TV}} \leq \|\mu - \nu\|_{\text{TV}},$$

let $\Delta := \mu - \nu$. Use the dual characterization of TV and Fubini's theorem,

$$\begin{aligned} \|K\mu - K\nu\|_{\text{TV}} &= \frac{1}{2} \sup_{\|f\|_\infty \leq 1} \left| \int f d(K\Delta) \right| \\ &= \frac{1}{2} \sup_{\|f\|_\infty \leq 1} \left| \int (Kf)(x) d\Delta(x) \right| \\ &\leq \frac{1}{2} \sup_{\|g\|_\infty \leq 1} \left| \int g(x) d\Delta(x) \right| = \|\mu - \nu\|_{\text{TV}}, \end{aligned}$$

where we set $g = Kf$, which still satisfies $\|g\|_\infty \leq 1$. Let $K := K_{t \rightarrow 0}$ and $p := Kp_t$. Applying the above inequality to q_t, p_t gives

$$\|Kq_t - Kp_t\|_{\text{TV}} \leq \|q_t - p_t\|_{\text{TV}}.$$

Thus, we obtain a contraction $\kappa \in [0, 1]$. $\square$

To sharpen this result for the fixed pair (q_t, p_t) , let $\Delta = q_t - p_t = \Delta^+ - \Delta^-$ be its Jordan decomposition. For a non-degenerate Gaussian reverse kernel, the propagated measures $K\Delta^+$ and $K\Delta^-$ have strictly positive densities with nonzero overlap. This overlap causes strict cancellation, so whenever $q_t \neq p_t$,

$$\|Kq_t - Kp_t\|_{\text{TV}} < \|q_t - p_t\|_{\text{TV}}.$$

Therefore, a pair-dependent coefficient $\kappa(q_t, p_t) \in [0, 1)$ exists.

F.2 Proposition 3.1

Fix a threshold τ and follow Sec. 3.2, then we have

$$J^* := J^*(\tau), \quad A_\tau := \bigcup_{j \in J^*} B_j, \quad \alpha := \alpha(\tau) = p_t(A_\tau), \quad \beta := \beta(\tau) = q_t(A_\tau).$$

Recall the selected mixtures and weights

$$q_t^c = \sum_{j \in J^*} \tilde{\pi}_j q_{t|j}, \quad \tilde{\pi}_j = \frac{\hat{\pi}_j}{\beta}, \quad p_t^c = \sum_{j \in J^*} \bar{\pi}_j p_{t|j}, \quad \bar{\pi}_j = \frac{\pi_j}{\alpha}.$$

Proposition 3.1 (Selective Reverse Error Bound). *Under the DC-structure in Definition 3.1 and the batch construction described above, for any threshold τ the deviation between the selectively corrected reverse distribution $K_{t \rightarrow 0} q_t^c(\tau)$ and the true distribution $p = K_{t \rightarrow 0} p_t$ admits an upper bound of the form*

$$\|K_{t \rightarrow 0} q_t^c(\tau) - p\|_{\text{TV}} \leq f(\alpha(\tau), \beta(\tau), (\pi_j, \hat{\pi}_j)_{j \in J^*(\tau)}, \sup_{j \in J^*(\tau)} \text{TV}(q_{t|j}, p_{t|j})), \quad (11)$$

where $f(\cdot)$ is an explicit function of the true coverage $\alpha(\tau)$, the model coverage $\beta(\tau)$, the selected batch weights, and the conditional discrepancies.

Proof. Conditioning p_t on A_τ drops exactly $1 - \alpha$ true mass, hence **coverage term** is

$$\|p_t^c - p_t\|_{\text{TV}} = 1 - \alpha. \quad (14)$$

Add and subtract $\sum_{j \in J^*} \tilde{\pi}_j p_{t|j}$ and use the triangle inequality to obtain **selected-region term**:

$$\|q_t^c - p_t^c\|_{\text{TV}} \leq \underbrace{\left\| \sum_{j \in J^*} \tilde{\pi}_j (q_{t|j} - p_{t|j}) \right\|_{\text{TV}}}_{\text{component error}} + \underbrace{\left\| \sum_{j \in J^*} (\tilde{\pi}_j - \bar{\pi}_j) p_{t|j} \right\|_{\text{TV}}}_{\text{weight mismatch}}. \quad (15)$$

For the **component term**, TV is convex:

$$\left\| \sum_j \tilde{\pi}_j (q_{t|j} - p_{t|j}) \right\|_{\text{TV}} \leq \sum_j \tilde{\pi}_j \|q_{t|j} - p_{t|j}\|_{\text{TV}}. \quad (16)$$

Now separate the shared and unshared parts of the two distributions. Let $c_j := \min\{\tilde{\pi}_j, \bar{\pi}_j\}$ and write $\tilde{\pi}_j = c_j + r_j$, $\bar{\pi}_j = c_j + s_j$ with $r_j, s_j \geq 0$ and $\sum_j r_j = \sum_j s_j = \frac{1}{2} \sum_j |\tilde{\pi}_j - \bar{\pi}_j|$. Then

$$\sum_j \tilde{\pi}_j (q_{t|j} - p_{t|j}) = \sum_j c_j (q_{t|j} - p_{t|j}) + \sum_j r_j q_{t|j} - \sum_j s_j p_{t|j}.$$

Using convexity,

$$\left\| \sum_j \tilde{\pi}_j (q_{t|j} - p_{t|j}) \right\|_{\text{TV}} \leq \sum_j c_j \|q_{t|j} - p_{t|j}\|_{\text{TV}} + \frac{1}{2} \sum_j |\tilde{\pi}_j - \bar{\pi}_j|. \quad (17)$$

For the weight term in Eq. (15), because $\sum_j (\tilde{\pi}_j - \bar{\pi}_j) = 0$,

$$\left\| \sum_{j \in J^*} (\tilde{\pi}_j - \bar{\pi}_j) p_{t|j} \right\|_{\text{TV}} \leq \frac{1}{2} \sum_{j \in J^*} |\tilde{\pi}_j - \bar{\pi}_j|. \quad (18)$$

Combining Eq. (17) and Eq. (18), and noting $\sum_j c_j \leq 1$, gives

$$\|q_t^c - p_t^c\|_{\text{TV}} \leq \sup_{j \in J^*} \|q_{t|j} - p_{t|j}\|_{\text{TV}} + \|\tilde{\pi} - \bar{\pi}\|_1. \quad (19)$$

Renormalize the weights,

$$\begin{aligned} \|\tilde{\pi} - \bar{\pi}\|_1 &= \sum_{j \in J^*} \left| \frac{\hat{\pi}_j}{\beta} - \frac{\pi_j}{\alpha} \right| = \frac{1}{\alpha\beta} \sum_{j \in J^*} |\alpha\hat{\pi}_j - \beta\pi_j| \\ &= \frac{1}{\alpha\beta} \sum_{j \in J^*} |\alpha(\hat{\pi}_j - \pi_j) + (\alpha - \beta)\pi_j| \leq \frac{1}{\alpha\beta} \left(\alpha \sum_{j \in J^*} |\hat{\pi}_j - \pi_j| + |\alpha - \beta| \sum_{j \in J^*} \pi_j \right) \\ &= \frac{\|\hat{\pi} - \pi\|_1}{\beta} + \frac{|\alpha - \beta|}{\beta} \leq \frac{2\|\hat{\pi} - \pi\|_1}{\beta} \leq \frac{2\|\hat{\pi} - \pi\|_1}{\min\{\alpha, \beta\}}. \end{aligned} \quad (20)$$

By symmetry (swapping α, β) we get $\|\tilde{\pi} - \bar{\pi}\|_1 \leq 2\|\hat{\pi} - \pi\|_1/\alpha$, so Eq. (20) is the uniform bound.

By the triangle inequality with Eq. (14) and Eq. (19), the final discrepancy at timestep t

$$\|q_t^c - p_t\|_{\text{TV}} \leq (1 - \alpha) + \|\tilde{\pi} - \bar{\pi}\|_1 + \sup_{j \in J^*} \|q_{t|j} - p_{t|j}\|_{\text{TV}}.$$

Using Eq. (20) we obtain the safe bound

$$\|q_t^c - p_t\|_{\text{TV}} \leq (1 - \alpha) + \frac{2\|\hat{\pi} - \pi\|_1}{\min\{\alpha, \beta\}} + \sup_{j \in J^*} \|q_{t|j} - p_{t|j}\|_{\text{TV}}. \quad (21)$$

Let $K_{t \rightarrow 0}$ be the ideal reverse kernel and $p := K_{t \rightarrow 0} p_t$. By Appendix F.1, for the fixed pair (q_t^c, p_t) there exists a pair-dependent coefficient $\kappa(q_t^c, p_t) \in [0, 1]$ such that

$$\|K_{t \rightarrow 0} q_t^c - K_{t \rightarrow 0} p_t\|_{\text{TV}} \leq \kappa(q_t^c, p_t) \|q_t^c - p_t\|_{\text{TV}}. \quad (22)$$

Please refer to Appendix F.3 for further details about the confidence bound and estimation error of α and β .

F.3 Confidence Bound

Even though the intermediate distributions are intractable, some quantities mentioned in Appendix F.2 can be empirically estimated from available samples of both the forward and reverse diffusion processes, and their estimation errors can be rigorously bounded. In diffusion models we can draw i.i.d. samples z_t from p_t and z'_t from q_t , we estimate

$$\hat{\alpha}(\tau) = \frac{1}{n_p} \sum_{i=1}^{n_p} \mathbf{1}\{z_t^{(i)} \in A_\tau\}, \quad \hat{\beta}(\tau) = \frac{1}{n_q} \sum_{i=1}^{n_q} \mathbf{1}\{z'_t{}^{(i)} \in A_\tau\}.$$

And by Hoeffding’s inequality,

$$\mathbb{P}(|\hat{\alpha} - \alpha| \leq \epsilon_\alpha) \geq 1 - \delta_\alpha, \quad \epsilon_\alpha = \sqrt{\frac{\log(2/\delta_\alpha)}{2n_p}},$$

and similarly for $\hat{\beta}$. We define lower confidence bounds

$$\alpha_L = \max\{\hat{\alpha} - \epsilon_\alpha, 0\}, \quad \beta_L = \max\{\hat{\beta} - \epsilon_\beta, 0\}.$$

For a fixed batch partition and threshold, define separate empirical batch-mass estimates from the forward- and reverse-process samples:

$$\pi_{\text{emp},j}^p = \frac{1}{n_p} \sum_{i=1}^{n_p} \mathbf{1}\{z_t^{(i)} \in B_j\}, \quad \pi_{\text{emp},j}^q = \frac{1}{n_q} \sum_{i=1}^{n_q} \mathbf{1}\{z'_t{}^{(i)} \in B_j\}.$$

Applying the multinomial L_1 concentration inequality separately gives, with probability at least $1 - \delta_p - \delta_q$,

$$\|\pi_{\text{emp}}^p - \pi\|_1 \leq U(n_p, \delta_p), \quad \|\pi_{\text{emp}}^q - \hat{\pi}\|_1 \leq U(n_q, \delta_q),$$

where

$$U(n, \delta) = \sqrt{\frac{2 \log((2^J - 2)/\delta)}{n}}.$$

Therefore, by the triangle inequality,

$$\|\hat{\pi} - \pi\|_1 \leq \|\pi_{\text{emp}}^q - \pi_{\text{emp}}^p\|_1 + U(n_q, \delta_q) + U(n_p, \delta_p).$$

Combining Eq. (21) with Eq. (22), substituting the relation above, and writing $\kappa := \kappa(q_t^c, p_t)$, gives

$$\|K_{t \rightarrow 0} q_t^c(\tau) - p\|_{\text{TV}} \leq \kappa \left[(1 - \alpha_L) + \frac{2(\|\pi_{\text{emp}}^q - \pi_{\text{emp}}^p\|_1 + U(n_q, \delta_q) + U(n_p, \delta_p))}{\min\{\alpha_L, \beta_L\}} + \sup_{j \in J^*(\tau)} \text{TV}(q_{t|j}, p_{t|j}) \right].$$

Provided that $\min\{\alpha_L, \beta_L\} > 0$, this bound holds with probability at least $1 - (\delta_\alpha + \delta_\beta + \delta_p + \delta_q)$. The coverage and batch-mass terms can be estimated from samples, while the conditional TV term remains a population quantity.

G Implementation Details of DIST

DIST is applied at an intermediate timestep t to correct the distribution q_t . Concretely, when applied to EDM [Hoogeboom et al., 2022] and GeoLDM [Xu et al., 2023], DIST firstly denoises samples from $t = 1000$ to $t = 300^3$, then duplicates and perturbs each sample 100 times to form batches $\{B_j\}_{j=1}^J$ under a radius- r constraint. From each batch, half of the elements are randomly selected as pilot subsets $\{B_j^{\text{sub}} \mid B_j^{\text{sub}} \subset B_j\}_{j=1}^J$, and pilot outcomes $s_j \in \mathbb{R}$ are evaluated based on the stability and validity of the final generated molecules. When applied to RADM [Ding and Hofmann, 2025], the settings are the same except for: the pilot subsets on QM9, which are fixed as 30% of the batch size; and the first-stage denoising on GEOM-Drugs, which terminates at $t = 200$.

G.1 DIST Inference Procedure

This section provides the pseudocode for DIST in Algorithm 1.

³All three baselines use a total of $T = 1000$ timesteps.

Algorithm 1 DIFFUSE AND STEER (DIST)

Inputs: neural network ε_θ , target number N , intermediate timestep t^* , threshold τ , pilot size $|B^{\text{sub}}|$, pilot score module Eval
Output: set of generated molecules $\mathbf{S}$
Initialize $\mathbf{S} \leftarrow \emptyset$
Sample initial noise batch $\{\mathbf{z}_T\} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
while $|\mathbf{S}| < N$ **do**
 for $t = T, T-1, \dots, t^*+1$ **do**
 Sample $\{\varepsilon\} = \{(\varepsilon_x, \varepsilon_h)\} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 Subtract center of mass from ε_x
 $\{\mathbf{z}_{t-1}\} \leftarrow \frac{1}{\sqrt{1-\beta_t}} \left(\{\mathbf{z}_t\} - \frac{\beta_t}{\sqrt{1-\alpha_t}} \varepsilon_\theta(\{\mathbf{z}_t\}, t) \right) + \rho_t \{\varepsilon\}$
 end for
 Duplicate and perturb each element in $\{\mathbf{z}_{t^*}\}$ to obtain local batches $\{\mathbf{z}_{t^*}^{(j,r)}\}_{j=1}^J$ {within the prescribed radius}
 Select pilot subsets $\{\mathbf{z}_{t^*}^{\text{sub},(j,r)}\}_{j=1}^J$ of size $|B^{\text{sub}}|$ per batch
 $\{s_j\}_{j=1}^J, \{\mathbf{z}_0^{\text{sub},(j,r)}\}_{j=1}^J \leftarrow \text{Eval}(\{\mathbf{z}_{t^*}^{\text{sub},(j,r)}\}_{j=1}^J)$
 $\mathbf{S} \leftarrow \mathbf{S} \cup \{\mathbf{z}_0^{\text{sub},(j,r)} \mid s_j \geq \tau\}$
 Define remaining high-score samples at t^* : $\{\mathbf{z}_{t^*}^{\text{rest},(j,r)}\} \leftarrow \{\mathbf{z}_{t^*}^{(j,r)} \mid s_j \geq \tau\} \setminus \{\mathbf{z}_{t^*}^{\text{sub},(j,r)} \mid s_j \geq \tau\}$
 for $t = t^*, t^*-1, \dots, 1$ **do**
 Sample $\{\varepsilon\} = \{(\varepsilon_x, \varepsilon_h)\} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 Subtract center of mass from ε_x
 $\{\mathbf{z}_{t-1}^{\text{rest},(j,r)}\} \leftarrow \frac{1}{\sqrt{1-\beta_t}} \left(\{\mathbf{z}_t^{\text{rest},(j,r)}\} - \frac{\beta_t}{\sqrt{1-\alpha_t}} \varepsilon_\theta(\{\mathbf{z}_t^{\text{rest},(j,r)}\}, t) \right) + \rho_t \{\varepsilon\}$
 end for
 $\mathbf{S} \leftarrow \mathbf{S} \cup \{\mathbf{z}_0^{\text{rest},(j,r)}\}$
end while

G.2 Pilot Evaluation Module

Algorithm 1 writes $\text{Eval}(\cdot)$ in a general form because DIST only requires a pilot score indicating whether a local intermediate batch is likely to return to a valid distributional region. This design allows future work to use other diagnostics, such as self-consistency, ensemble uncertainty, or domain-specific constraints. In our experiments, we use a concrete chemistry-based instantiation: Eval runs the remaining reverse process from t^* to 0 on the pilot subset, converts the outputs into molecules, and computes an RDKit-based score. For clarity, we provide the pseudocode in Algorithm 2.

Algorithm 2 Instantiation of Pilot Evaluation Module

Inputs: pilot subsets $\{\mathbf{z}_{t^*}^{\text{sub},(j,r)}\}_{j=1}^J$, neural network ε_θ , scoring function Score
Output: batch scores $\{s_j\}_{j=1}^J$ and pilot outputs $\{\mathbf{z}_0^{\text{sub},(j,r)}\}_{j=1}^J$
for $t = t^*, t^*-1, \dots, 1$ **do**
 Sample $\{\varepsilon\} = \{(\varepsilon_x, \varepsilon_h)\} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 Subtract center of mass from ε_x
 $\{\mathbf{z}_{t-1}^{\text{sub},(j,r)}\} \leftarrow \frac{1}{\sqrt{1-\beta_t}} \left(\{\mathbf{z}_t^{\text{sub},(j,r)}\} - \frac{\beta_t}{\sqrt{1-\alpha_t}} \varepsilon_\theta(\{\mathbf{z}_t^{\text{sub},(j,r)}\}, t) \right) + \rho_t \{\varepsilon\}$
end for
 $\{s_j\}_{j=1}^J \leftarrow \text{Score}(\{\mathbf{z}_0^{\text{sub},(j,r)}\}_{j=1}^J)$ {One scalar score per batch}
return $\{s_j\}_{j=1}^J, \{\mathbf{z}_0^{\text{sub},(j,r)}\}_{j=1}^J$

H Efficiency Quantification of DIST

H.1 Theoretical Timestep Efficiency

In this section, we quantify the expected computational cost of the DIST. Let T denote the total number of timesteps in standard inference, and let $t < T$ be the intermediate timestep at which we form the candidate pool. Each candidate is then duplicated into batches of size $|B|$, and only a fraction r_c are retained after pilot evaluation, which incurs an additional cost C_{pilot} per sample. Analytically, the expected cost for a generated sample will be:

$$\mathbb{E}[\text{cost}] = \left(\frac{1}{r_c} - 1\right) \left[\frac{T-t}{|B|} + C_{\text{pilot}}\right] + \left[\frac{T-t}{|B|} + t\right]. \quad (23)$$

The first term accounts for the discarded reverse inference steps, while the second term corresponds to the trajectory for selected batches requiring full reverse inference from t to 0. We now refine the cost expression by modeling the pilot evaluation as proportional to the selection timestep. Specifically, we set

$$C_{\text{pilot}} = \gamma t, \quad \gamma = \frac{|B^{\text{sub}}|}{|B|}, \quad (24)$$

where γ denotes the proportion of elements in a batch used for pilot evaluation (see Sec. 3.2). Substituting this into the expected cost gives

$$\mathbb{E}[\text{cost}] = \left(1 - \gamma + \frac{\gamma}{r_c}\right)t + \frac{T-t}{|B|r_c}. \quad (25)$$

Importantly, the expected cost depends on r_c , which is correlated with the final generation quality. However, in practice, the total cost is usually much smaller than T . For example, in the EDM+DIST experiment on QM9 (as shown in Table 2), the settings are $T = 1000$, $|B| = 100$, and $\gamma = 0.5$. The empirical estimate of r_c is near 0.39, leading to an empirical cost of 556.1 steps on average. For comparison, a score-matched post-hoc selector must fully denoise all candidates before retaining a fraction r_c . Its expected cost is therefore T/r_c timesteps per retained sample, excluding the small scoring overhead. With $T = 1000$ and $r_c \approx 0.39$, this requires approximately 2564 timesteps per retained sample, compared with 556.1 for DIST. DIST is more efficient because it shares computation before timestep t and stops discarded regions before completing their reverse trajectories.

Please refer to Algorithm 1 for detailed implementation of DIST and Appendix H.2 for end-to-end wall-clock usage for efficiency.

H.2 Wall-Clock Efficiency

In this section, we provide the end-to-end time efficiency of DIST using wall-clock runtime. All experiments are conducted on a single NVIDIA RTX A6000 GPU with CUDA 12.4. To ensure a fair comparison and remain consistent with prior work [Hooeboom et al., 2022, Xu et al., 2023, Ding and Hofmann, 2025], we follow the standard setting of generating 10,000 molecules with a fixed batch size of 100 (see Sec. 4.1).

To better explain the efficiency mechanism and justify the reproducibility of DIST, we present the pseudocode in Algorithm 1. For consistency with the theoretical foundation in Sec. 3.2 and to illustrate the parallelism of the data flow, the inference is performed in a batch-wise manner. In existing works [Hooeboom et al., 2022, Song et al., 2024, Xu et al., 2023, Ding and Hofmann, 2025], diffusion models generate molecules batch by batch, whereas DIST introduces a novel efficient paradigm that constructs an intermediate model distribution and explicitly evaluates its deviation from the true distribution. This selective correction is the source of the computational savings.

The iterative Markovian denoising procedure is the main bottleneck for speeding up inference in diffusion models [Frans et al., 2025], so the consumption of GPU-intensive timesteps is a key quantity to consider. In addition to the timestep-based analysis in Table 3 and Appendix H.1, we also report the end-to-end wall-clock runtime for DIST combined with the baseline models. As shown in Table 7, DIST consistently improves the efficiency of the backbone methods, in line with the timestep-based results in Table 3 and the theoretical analysis in Appendix H.1.

We would like to clarify that **DIST is designed as a corrective sampling method for stabilizing molecular generation, with its efficiency improvement being an advantageous side effect.** As such, DIST is not directly comparable to diffusion-model acceleration techniques.

Table 7: Wall-clock end-to-end consumption of different sampling strategies for generating 10,000 molecules on QM9. The experimental setting is identical to Table 2, using a fixed batch size of 100 and an intermediate timestep $t = 300$. ‘Count’ indicates how many attempts are required to produce 10,000 molecules (e.g., running inference with batch size 100 for 100 iterations). For models without DIST, the stages ‘Diffuse’ and ‘Steer’ correspond to the standard inference procedures $T \rightarrow t$ and $t \rightarrow 0$, respectively. The ‘Total’ column is reported in minutes and seconds (mm:ss.ss). The better strategy and its total wall-clock consumption are highlighted in **bold**.

Model	Diffuse		Duplication		Pilot Evaluation		Steer		Total ↓
	Time (s)	Count	Time (s)	Count	Time (s)	Count	Time (s)	Count	
EDM	60.935	100	–	–	–	–	26.115	100	145:07.11
EDM+DIST	60.933	3	0.0009	258	27.0525	129	26.115	50	82:59.45
GeoLDM	58.1163	100	–	–	–	–	25.0704	100	138:39.72
GeoLDM+DIST	58.1163	3	0.0008	220	25.9830	110	25.0705	50	71:26.91
RADM	26.4737	100	–	–	–	–	11.3250	100	63:00.53
RADM+DIST	26.4742	3	0.0008	219	11.9224	66	11.4118	70	27:47.10

I Supplementary Ablation Study

This section provides additional ablation studies for the hyperparameters of DIST, including the batch score threshold τ , the intermediate timestep t , and the perturbation intensity used in the radius- r assumption, each introduced in Sec. 3.2. For each ablation experiment, we vary only a single hyperparameter of DIST while keeping all other settings fixed to the configuration of DIST strengthened EDM [Hooeboom et al., 2022] as described in Sec. 4. This ensures that every sub-experiment isolates the effect of one hyperparameter change.

As discussed in Sec. 3.2, DIST operates purely as an inference-time corrective module and does not require retraining or modification of the backbone diffusion model. Because inference is substantially cheaper than training (e.g., in DDPM [Ho et al., 2020], training takes 10.6 hours on 8 V100 GPUs while generating 256 CIFAR10 samples takes only 17 seconds), **hyperparameter search is lightweight in practice**. Importantly, the results below show that DIST performs **consistently well across a broad range of hyperparameters**, indicating that the method is not overly sensitive to tuning.

I.1 Batch Score Threshold

Given a threshold τ , DIST selects batches according to whether their pilot scores exceed this value, which determines the coverage of both the true distribution and the model distribution. Larger thresholds restrict selection to batches that are more likely to correspond to valid high-density regions, whereas smaller thresholds broaden the selected region and capture more mass, while at the cost of admitting samples that deviate from the true distribution (see Sec. 3.2).

The results are reported in Table 8. Although a larger τ improves stability metrics, the admissible region becomes overly constrained and may harm overall performance. Moreover, the retained ratio r_c (see Appendix H.1) drops from 71% to 32%, indicating that a larger portion of batches is discarded, which in turn reduces the sampling efficiency. Across all tested values, DIST consistently outperforms the EDM [Hooeboom et al., 2022], demonstrating robustness to the choice of τ .

Table 8: Ablation on batch score threshold τ for molecule stability on QM9.

τ	Atom Sta (%)	Mol Sta (%)	Valid (%)	Valid×Unique (%)
EDM	98.7	82.0	91.9	90.7
0.82	98.9	87.8	95.4	92.4
0.84	99.1	88.2	96.6	93.9
0.86	99.2	89.9	96.9	94.1
0.88	99.2	90.2	96.8	93.2

I.2 Intermediate Timestep

The intermediate timestep t determines when the corrective selection is applied. If t is too large, forward noise destroys the DC-structure; if too small, the candidate pool does not adequately represent the intermediate distribution (see Sec. 3.2).

As shown in Table 9, smaller t values correct the distribution later in the reverse chain, improving stability metrics, while excessively small values may reduce uniqueness due to extended shared diffusing. Even without intentional tuning, tested t values yield strong improvements over the EDM baseline, indicating that DIST remains effective across a broad range.

Table 9: Ablation on intermediate timestep t on QM9.

t	Atom Sta (%)	Mol Sta (%)	Valid (%)	Valid×Unique (%)
EDM	98.7	82.0	91.9	90.7
200	99.2	90.2	96.8	92.5
300	99.2	89.9	96.9	94.1
400	99.2	89.7	96.0	94.0
500	99.1	89.3	95.4	92.1

I.3 Perturbation Intensity

Following Definition 3.1 and the discussion in Sec. 3.2, the space is partitioned into radius- r batches $\{B_j\}_{j=1}^J$, each serving as a local region around peaks. After obtaining the candidate pool at timestep t , these batches are optionally perturbed with Gaussian noise of scale λ .

Table 10 shows that DIST is highly robust to a small perturbation intensity: tested λ values produce strong performance, and even $\lambda = 0$ (pure duplication without perturbation) already offers substantial improvements. This indicates that DIST does not rely on fine-grained tuning of perturbation intensity.

Table 10: Ablation on perturbation intensity λ on QM9.

λ	Atom Sta (%)	Mol Sta (%)	Valid (%)	Valid×Unique (%)
EDM	98.7	82.0	91.9	90.7
0	99.2	89.9	96.9	94.1
0.05	99.2	89.7	95.9	93.1
0.1	99.1	90.1	95.9	92.8
0.2	99.3	90.3	96.2	92.7